Estimating Tropical Cyclone Intensity via Deep Learning and Storm Morphology Recognition From Satellite Imagery

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***Abstract*—It is well-established that the intensity of tropical cyclones (TCs) is highly associated with their structure. This understanding allows us to utilize geostationary satellite cloud products for the recognition of storm morphology, thereby enabling the accurate estimation of TC intensity. This study adapts the Swin-Unet architecture as a backbone model for an objective deep learning (DL)-based TC intensity estimation method over the Western North Pacific. The model incorporates several key components, including the self-attention mechanism, the shift-window mechanism, and the Unet structure. In addition, the model innovatively introduces the rotation index and the dispersion index as part of the loss function to characterize storm morphology and perform more comprehensive feature extraction from spatiotemporal data. The model's input in the study includes five cloud products from the Fengyun series geostationary satellites: sectional image (SEC), cloud top temperature (CTT), temperature of the brightness blackbody (TBB), precipitation estimation (PRE), and humidity profile derived from cloud analysis (HPF). Results show that the model obtains an exceptionally low mean absolute error (MAE) of 3.71 m/s and a root mean square error (RMSE) of 5.05 m/s. Furthermore, the ablation study (component-impact analysis) quantifies the contribution of the rotation index and the dispersion index, which enhance the model's estimation performance to some extent. Finally, through an analysis of feature importance across the five cloud products, HPF, CTT, and TBB received higher importance scores, indicating the model concentrates on the thermodynamic and dynamic features that are strongly associated with TC convective activities. This study is expected to provide technical support for real-time TC intensity estimation in coastal regions and contribute to disaster warning systems.**

***Index Terms*—Deep learning (DL), intensity, morphology recognition, satellite imagery, tropical cyclone (TC).**

I. Introduction

TROPICAL cyclone (TC) is a low-pressure vortex system forming over tropical or subtropical oceans, usually accompanied by strong winds, heavy rainfalls, and storm surges, causing enormous losses of life and property in coastal areas [1], [2], [3]. The TC intensity is usually measured by the maximum surface wind speed or the central minimum pressure of a cyclone, acting as an important factor in determining the severity of TC disaster risk [4]. Therefore, timely, accurate, and objective estimation of TC intensity is of vital importance for disaster prevention and mitigation, as well as for management decision-making. However, due to the intrinsic complexity of convection and thermodynamics of the atmosphere that is related to TC activities [5], [6], [7], TC intensity estimation has long been a challenge in the past few decades [8], [9], [10], [11].

The widely used method in estimating TC intensity is the Dvorak technique (DT). TCs with different intensities usually exhibit distinct cloud patterns, and DT reckons TC intensity [12] by examining the characteristics of cloud pattern variations with a focus on TC eyes, storm eyewalls, and TC outer spiral rainbands. However, this technique highly relies on expert human analysts and the quality of satellite imagery [13], [14]. Therefore, the DT can be considered a form of manual intervention technology with the subjectivity of the assessment. In recent decades, increasingly optimized objective techniques to estimate TC intensity using visible and infrared satellite data have been developed to complement the manual DT process, such as the objective DT [15], the advanced objective DT [16], the advanced DT (ADT) [17], and the deviation angle variance technique [18], [19]. The success of these objective techniques can be attributed to the association between thermodynamic and dynamic factors of TC intensity changes and observable cloud patterns. However, they still have an inherent weakness. For example, they often underestimate the intensity of small TCs and large TCs; they lack consideration of the environmental winds. Moreover, the subjective rules and constraints may lead to an insufficient response to the rapid intensification of TCs [6].

Numerous studies show that various aspects of a TC structure, such as its size, fullness, the rotation of its spiral rainband, and the asymmetric structure in the eyewall, may have certain impacts on the intensity changes, including rapid intensification [20], [21], [22], [23], [24], [25]. However, other studies suggest that the relationship between the size and intensity is complex and nonlinear [26], [27], [28], [29]. The TC size (usually measured by the radius of maximum wind), particularly its initial size [30], [31], may respond to TC external environments and internal dynamics. Therefore, in many cases, it can help maintain the moisture structure,

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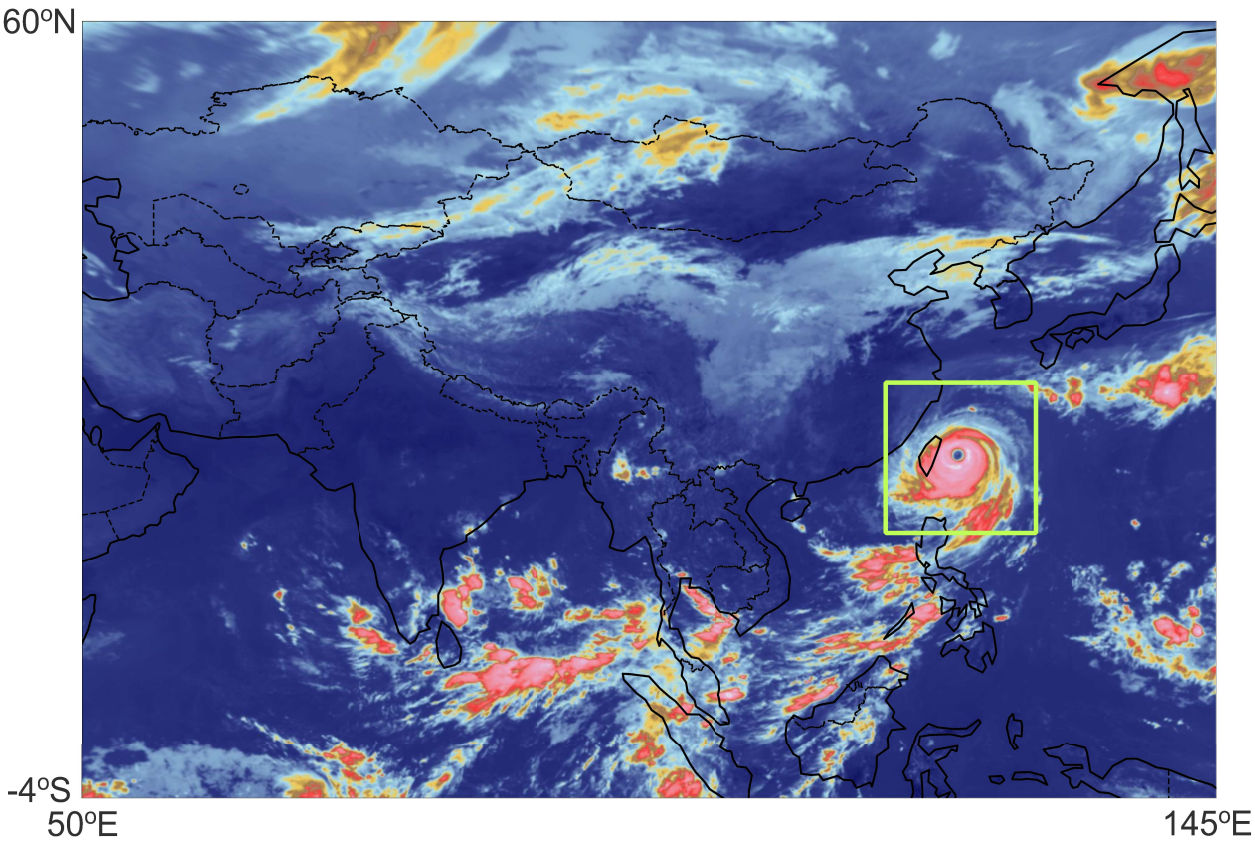


Fig. 1. Coverage area of the FY-2 meteorological satellite. The figure shows the sectional image (SEC) captured by FY-2G satellite on September 28, 2015, at 03:00 AM, during the super typhoon "Dujuan."

which supports the development of intensity. However, this does not directly translate to a higher intensity. Guo and Tan [25] defined "TC fullness" to quantify the size configuration of TCs and suggested that both the TC intensity and intensification are modulated by TC size configuration. Compared to the TC size, the relationship between the TC intensity and the rotation of the TC's spiral rainband and the symmetrical structure is clearer. Wang [32] proved that there is a strong connection between the vortex Rossby waves and the polygonal eyewall structure and its cyclonic rotation. Chao et al. [21] investigated the rainband rotation speeds from satellite imagery and demonstrated that it is highly correlated with the TC intensity. It is also assumed that the correlation can be further improved by considering the outflow speed of TCs. Meanwhile, [33] suggested that the initial asymmetric structure in a storm's vortex can enhance or suppress the TC intensity in the environment of vertical wind shear. Therefore, although the relationship between TC intensity and structure is still difficult to express with precise statistical formulas, it is multifaceted and involves complex interactions between TC's external environments and internal dynamics. This relationship is also a key focus in estimating TC intensity.

With the development of data collection technology such as meteorological satellites and radars, and the high-performance parallel computing technology for graphics processing units (GPUs), the integration of deep learning (DL) with satellite imagery for TC intensity estimation has garnered substantial attention and acclaim. Compared with traditional statistical regression techniques, the DL technique boasts robust self-learning and nonlinear-mapping ability, rendering it suited for tackling the high-dimensional satellite imagery data. For instance, these studies [6], [8], [34], [35], [36], [37], [38], [39], [40], [41] introduced convolutional neural network (CNN)-based method for estimating TC intensity, with satellite imagery sourced from diverse bands/channels serving as the primary inputs for the model. Typically, the CNN-based approach demonstrates a lower estimation error compared to DT-based families, indicating a higher estimation accuracy. From the perspective of computer vision, the CNN method excavates complex spatiotemporal patterns and nonlinear features of satellite imagery through operations such as local receptive field, convolution, weight-bias sharing, and pooling, thereby establishing the relationship between morphology recognition of satellite imagery and TC intensity, which facilitates accurate TC intensity estimation. Furthermore, numerous studies also documented the progress and achievements in estimating TC intensity using other DL methods alongside statistical techniques. Studies such as [9] introduced a deep multisource attention network and integrated it with a Kalman filter. Tong et al. [42] presented a convolutional long short-term memory (LSTM) network that is capable of extracting both temporal correlations and parameter-based dependencies from the input data. Zhang et al. [43] proposed the "TC-Pred" model, which incorporates a convolutional transformer module to alleviate the long-term dependence decay problem

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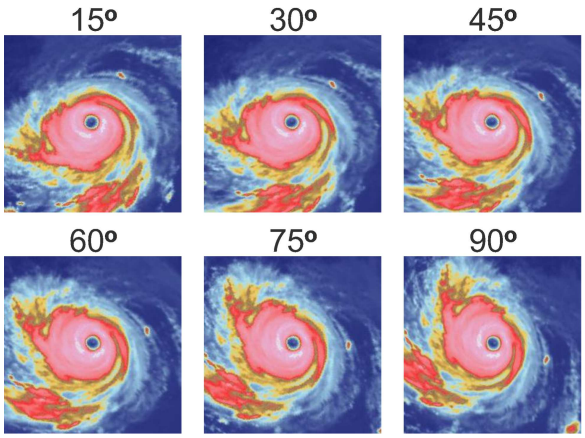


Fig. 2. Example of data augmentation under the sectional image (SEC) during the super typhoon "Dujuan" on September 28, 2015, at 03:00 AM.

from environmental variables. The above studies reported the superiority of the DL-based method in estimating TC intensity, especially for specific regions and periods.

However, the primary shortcoming of employing DL methods (particularly CNN-based networks) and satellite imagery for estimating TC intensity is the lack of detailed depictions of the TC's structure, which consequently restricts the accuracy. It is well known that TC intensity is related to the structure of storms, including their size, fullness, rotation in spiral rainbands, and the symmetric cloud pattern. Consequently, most DL methods are sensitive to deformations, rotations, or scaling when analyzing satellite imagery. The shape, size, and position of clouds can change over time and with varying synoptic conditions, which may pose challenges for DL models in storm morphology recognition. According to the above analysis, this study aims to reduce the error in TC intensity estimation by incorporating storm morphology indices derived from satellite imagery into the DL network. This modification enables the network to comprehensively analyze the storm's morphological features, thereby improving the accuracy.

The rest of this article is organized as follows. Section II presents a description of the data sources. Section III describes the method of the DL model, the model development, and the experiment design. Section IV provides the main experimental results. The discussion and conclusion are included in Sections V and VI, respectively.

II. Data Source

A. Fengyun-2 Geostationary Satellite Cloud Products

The FY-2 meteorological satellite is China's first generation of geostationary orbit meteorological satellites, which complements the polar-orbiting meteorological satellites to form China's meteorological satellite application system. It furnishes a vast array of global and regional Earth observations for utilization across various domains, including meteorology, oceanography, forestry, agriculture, civil aviation, and military affairs [44]. Specifically, FY-2F and FY-2G, which were launched in 2012 and 2014, respectively, stand out among them. The scanning radiometers on these two satellites have one visible channel and four infrared channels, offering more flexible, high temporal resolution scanning capabilities for specific areas, enabling focused observations of TC over the northwest Pacific (50°E ~ 145°E, 4°S ~ 60°N; see Fig. 1). In this study, we select five cloud products from FY-2F and FY-2G with a period ranging from 2015 to 2023, including sectional image (SEC), cloud top temperature (CTT), temperature of brightness blackbody (TBB), precipitation estimation (PRE), and humidity profile derived from cloud analysis (HPF). Their temporal resolutions are all 1 h. The spatial resolution of SEC is 5 km, whereas the spatial resolution of the other four products is 10 km. They all cover the northwest Pacific basin where the majority of TCs are active. To a certain extent, these five products directly reflect the convective activities, TC spiral rainbands, cloud pattern distribution, and precipitation patterns surrounding TCs, providing valuable insights into TC structures.

B. TC Data

The TC data are sourced from Zhejiang's Typhoon Track Monitoring System, which acquires its Best-Track records directly from the China Meteorological Administration's National Meteorological Center. This system provides data for pelagic TCs every 2 h and for inshore TCs every 1 h. It contains detailed TC tracks and their associated time, longitude, latitude, minimum central pressure, maximum sustained surface wind speed (2 min average), direction of movement, speed of movement, and landfall location over the northwest Pacific basin. We select a total of 225 TC cases (comprising 12705 storm records) from nine TC seasons spanning from 2015 to 2023, which coincides with the period of the above satellite cloud products. We match each TC record with corresponding satellite imagery based on precise spatiotemporal coordinates (timestamp and latitude/longitude), ensuring unambiguous correspondence between every TC record and its associated satellite data. Based on the TC intensity classification criteria by the China Meteorological Administration, there are six types of TC: tropical depression (TD), tropical storm (TS), strong tropical storm (STS), typhoon (TY), strong typhoon (STY), and super STY (SSTY). In this study, we exclude TD records with wind speeds less than 17.2 m/s. The wind speed, which is also used as the training label, is considered TC intensity.

C. Data Preprocessing

Cloud products for 1280 × 1280 km areas centered on geographical positions in the storm record are extracted from full disk satellite imagery. Note that SEC is downsampled to match the 10-km spatial resolution of the other four products. In addition, due to the large proportion of TS and STS compared to the smaller proportion of TY and STY, we utilize the "data augmentation" technique [6], [34], [38] to not only increase the number of original samples but also alleviate the imbalance among them. Specifically, data augmentation is achieved by rotating derived storm-centered satellite images at various angles (see Fig. 2). Meanwhile, to better capture a wider range of features, including edges, textures, and colors, we load the images in the three-channel "RGB" format rather

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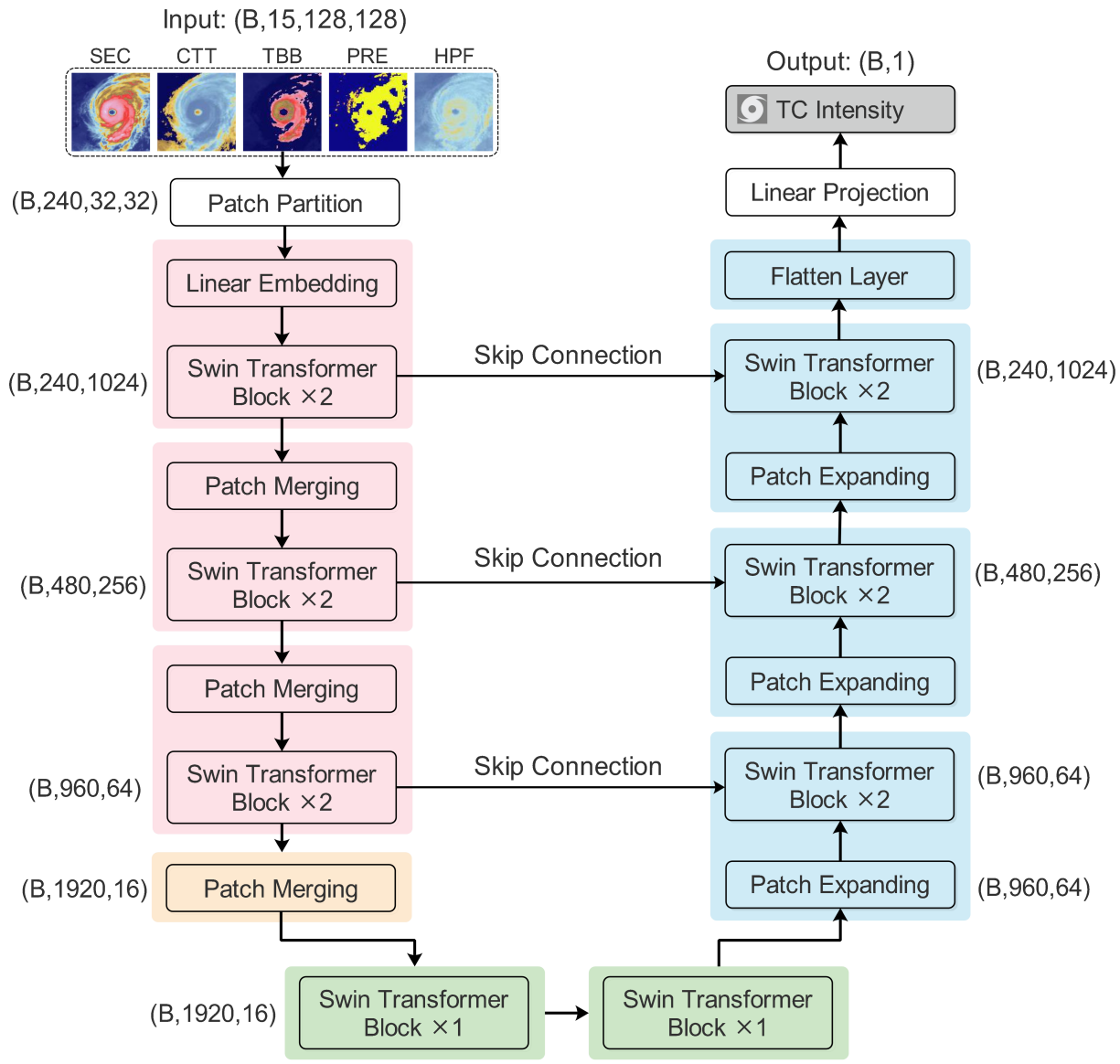


Fig. 3. Overall framework of the Swin-Unet model. Note that "B" denotes batch size, representing the number of samples processed per iteration during the Swin-Unet model training.

than the gray-scale format from the original satellite imagery. Therefore, an array with the size of (15 × 128 × 128) is cropped from 5 satellite cloud products according to each storm record. Approximately 790 outlier samples are eliminated due to equipment malfunctions in data acquisition, unstable satellite orbits, and the impact of eclipse periods, ultimately retaining 64647 valid samples.

III. Method and Model

A. Swin-Unet

Universally, the U-Net, as a CNN-based architecture [45], struggles with capturing global and long-range dependencies due to the locality of convolutional operations. In contrast, the transformer model, or specifically the vision transformer (ViT) model, is adept at capturing long-range dependencies in sequential data through its self-attention mechanism [46], [47], which enables the models to have potential advantages in processing global features of images. Therefore, the Swin-Unet model, proposed by Cao et al. [48], is a pure ViT architecture inspired by the U-Net architecture; it incorporates the respective advantages of both architectures, making it suitable for tasks such as classification, segmentation, and regression of satellite imagery. In this study, the Swin-Unet model mainly consists of several modules: patch partition, linear embedding, Swin transformer block, patch merging, patch expanding, flatten layer, linear projection, and skip connection (see Fig. 3).

First, the five satellite images are segmented into several smaller patches with a patch size of 4 × 4 using the "patch partition" module, thereby transforming these training inputs into a sequence of embedding vectors. Note that the patches are segmented from the imagery in each of the three channels, shifting the total dimensionality from 15 × 128 × 128 to 240 × 32 × 32. Second, a "linear embedding" module is applied to the embedding vectors, mapping them into 2-D word vectors,

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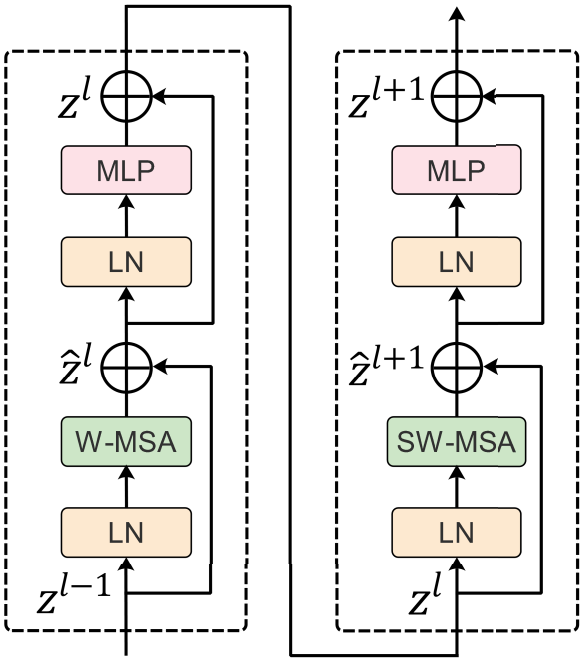


Fig. 4. Architecture of two successive Swin transformer blocks.

resulting in a sequence of patch tokens. Third, these patch tokens then enter the "Swin transformer block" for feature mining and extraction. The Swin transformer block is also the core module of the model. Fig. 4 shows the architecture of two consecutive Swin transformer blocks.

The Swin transformer introduces a "shift window" mechanism (SW-MSA) as a replacement for the conventional multihead self-attention (W-MSA) mechanism [46], [49]; it also consists of multilayer perceptron (MLP) and layer-norm (LN) layers. Through the above processes, the Swin transformer effectively reduces computational complexity, enhances cross-window information interaction capabilities, and better addresses boundary effects and information loss issues. Utilizing the shifted window partitioning method, the two consecutive Swin transformer blocks are represented by the following equations:

$$\hat{Z}^l = W - \mathrm{MSA}\left(\mathrm{LN}\left(Z^{l-1}\right)\right) + Z^{l-1} \tag{1}$$

$$Z^l = \mathrm{MLP}\left(\mathrm{LN}\left(\hat{Z}^l\right)\right) + \hat{Z}^l \tag{2}$$

$$\hat{Z}^{l+1} = \mathrm{SW} - \mathrm{MSA}\left(\mathrm{LN}\left(Z^l\right)\right) + Z^l \tag{3}$$

$$Z^{l+1} = \mathrm{MLP}\left(\mathrm{LN}\left(\hat{Z}^{l+1}\right)\right) + \hat{Z}^{l+1} \tag{4}$$

where $\hat{Z}^l$ and Z^l represent the output features of the (S)W-MSA module and the MLP module in the lth block, respectively. Furthermore, following the precedents set by [46] and [47], the self-attention mechanism can be computed as follows:

$$\mathrm{Attention}\,(Q, K, V) = \mathrm{SoftMax}\left(QK^T / \sqrt{d} + B\right) V \tag{5}$$

where $Q, K, V \in R^{M^2 \times d}$ denote the query, key, and value matrices, respectively; d is the dimension of query and key; and M^2 is the number of patches in a shift window. Suppose that $\hat{Z} \in R^{(2M-1)\times(2M-1)}$ is a parameterized bias matrix, so the values in B are taken from $\hat{B}$.

In addition, the patch merging module is applied to reduce the number of tokens and increase the feature dimension. This step typically follows the Swin transformer block. In contrast, the patch expanding module is used to expand the resolution of the input features and reduce the feature dimension; it also follows after the Swin transformer block. Overall, the Swin-Unet has a similar structure to Unet, featuring an encoder on the left part for downsampling and feature extraction, and a decoder on the right part for upsampling and feature fusion. The encoder and the decoder are connected through a "skip connection" to maximize the utilization of feature information across different layers of the network. Finally, after configuring the flattened layer and the linear projection layer, the network yields a TC intensity, which is a scalar value.

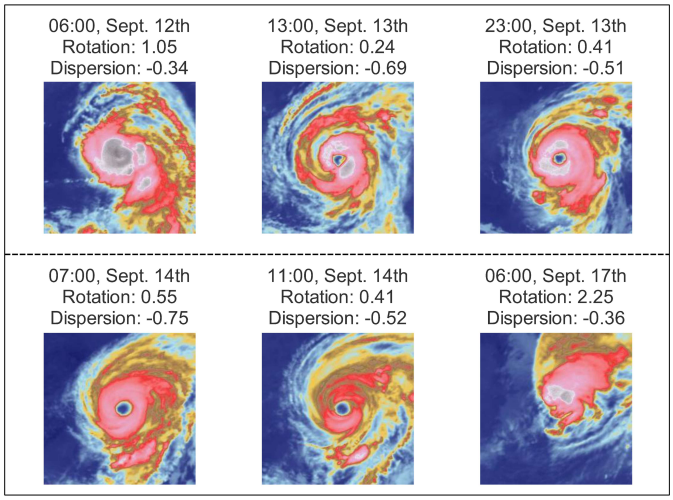


Fig. 5. Dispersion index and rotation index of Super Typhoon "Talim" (No. 201718) in 2017.

B. Loss Function

As mentioned above, TC intensity is related to the structure of the storm, including its size, fullness, rotation in spiral rainbands, and the symmetric cloud pattern. Therefore, this study defines a loss function that can not only quantify the loss of training on TC intensity but also TC structure. It can be given by the following formula:

$$\mathrm{Loss} = L_{\mathrm{MAE}} + L_R + L_D \tag{6}$$

where L_{MAE} represents the mean absolute error (MAE) function, used to quantify the loss during training concerning TC intensity. L_R and L_D, on the other hand, correspond to the rotation index and the dispersion index, respectively, and are used to quantify the loss incurred during training for the TC structure

$$L_{\mathrm{MAE}} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{7}$$

where y_i is the TC maximum wind speed ($V_{\max}$) derived from the Best-Track report dataset, $\hat{y}_i$ is the estimated TC intensity, and n is the number of TC samples.

The calculation of L_R is given as follows. First, we define the contrast between the pixels in the central region of an image $I(x, y)$ and the pixels in other regions of the image

$$C = |\mu_c - \mu_o| \tag{8}$$

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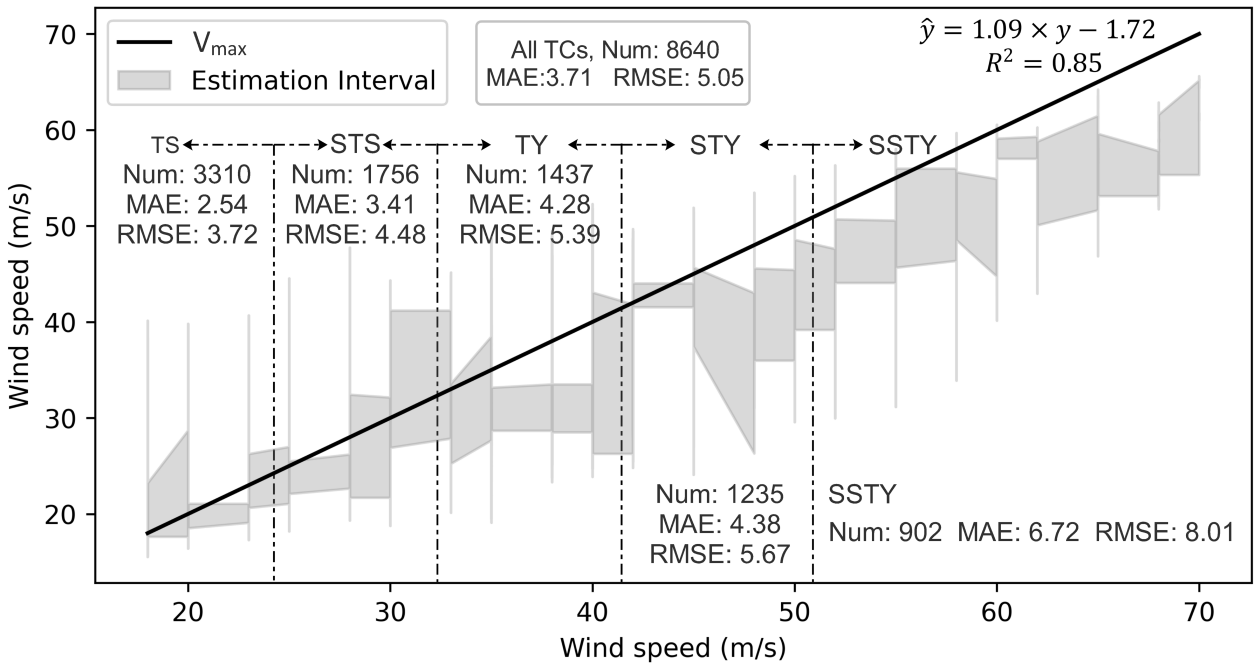


Fig. 6. Overall estimated results of TC intensity (unit: m/s) during 2015–2023. The solid black line represents the TC maximum wind speed ($V_{\max}$) derived from the Best-Track dataset, and the shaded gray area indicates the corresponding estimation intervals from the Swin-Unet model.

where μ_c and μ_o are the mean values of the pixels in the central region and other regions of the image, respectively. Then, we employ the Fourier transform to analyze the rotational symmetry of the image

$$F(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} I(x,y)\, e^{-2\pi i(ux+vy)} dxdy \tag{9}$$

where $F(u,v)$ denotes the expression of the image in the frequency domain. Thus, we have

$$M(u,v) = |F(u,v)| = \sqrt{[R(u,v)]^2 + [I(u,v)]^2} \tag{10}$$

where $M(u,v)$ stands for the magnitude spectrum of $F(u,v)$, and $R(u,v)$ and $I(u,v)$ are the real and imaginary parts of $F(u,v)$. Next, perform a polar coordinate transformation on $M(u,v)$; therefore,

$$R = \frac{1}{2\pi}\int_0^{2\pi} \sigma(M(r,\theta))\, d\theta \tag{11}$$

where $M(r,\theta)$ is the mean value along the radius r at angle θ (polar coordinate). $\sigma(M(r,\theta))$ corresponds to the standard deviation of $M(r,\theta)$. Last, by combining C and R, a comprehensive rotation index L_R is obtained

$$L_R = \alpha \cdot C + \beta \cdot R \tag{12}$$

where α and β are weight coefficients. The rotation index L_R illustrates the degree of rotation of each pixel in the image (see Fig. 5). The L_R is a "smaller is better" type of metric.

The calculation of L_D is given as follows. Given an image $I(x,y)$ with a size of $H \times W$, where the center coordinate of the image is (c_x, c_y), and the global mean value of the image is u, here, we define the weight W_I for each pixel value in the image

$$W_I = \frac{1}{\mu}\int_0^H \int_0^W \frac{I(x,y)}{d(x,y)} dxdy \tag{13}$$

$$d(x,y) = \sqrt{(x-c_x)^2 - \left(y-c_y\right)^2} \tag{14}$$

where $d(x,y)$ denotes the distance from the pixel (x,y) to the center pixel (c_x, c_y). Then, the entropy H of the image can be expressed by the following formula:

$$H = -\int_0^{255} p(i)\log(p(i))\, di \tag{15}$$

where $p(i)$ represents the probability density function of the pixel value i. Therefore, the expression for the dispersion index L_D is given as follows:

$$L_D = \frac{1+H}{W_I}. \tag{16}$$

The dispersion index L_D depicts the distribution uniformity and clustering of pixels in an image (see Fig. 5). The smaller the value of L_D, the better the training performance of the model.

IV. Results

A. Overall Results

To evaluate the estimated performance of the Swin-Unet model, we adopt the fivefold cross-validation method, which divides the dataset into a training set and an independent testing set. We partition the dataset based on TC cases, ensuring that data from the same TC are never included in both the testing and training datasets. For the cross-validation procedure, we select approximately 20% of TC cases as the independent testing set, while the remaining around 80% is used as the training and validation sets. The comprehensive results of these experiments are presented in Fig. 6. Intuitively, the estimated intensities are highly correlated with $V_{\max}$ with a coefficient of determination (R^2) of 0.85. The Swin-Unet model performs

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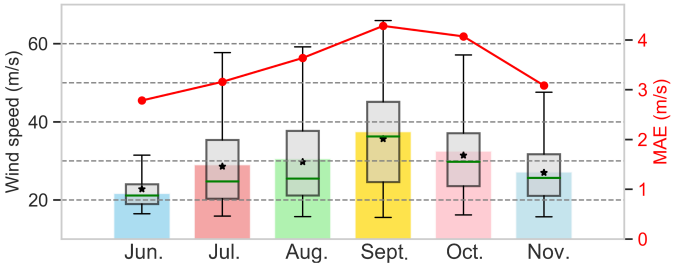


Fig. 7. Estimated results of TC intensity are from June to November. The bar chart displays the average $V_{\max}$, and the box-plot means shows the maximum, minimum, median (marked in *), 0.25 quantile, 0.75 quantiles, and the average value (green line) of the estimated TC intensity. The line chart (right Y-axis in red) depicts the MAE of estimated TC intensity.

well in estimating TS and STS, achieving MAEs of less than 4 m/s for both. However, the MAE gradually increases as the TC intensity grows stronger. Although SSTY makes up a small proportion of all TCs, its MAE has exceeded twice that of TS and STS. In terms of the estimation intervals, the model appears to exhibit a certain degree of overestimation for TS and STS, whereas it notably underestimates TY, STY, and SSTY. Especially, the model struggles to estimate the intensity within the range of 60 ~ 70 m/s, and it fails to estimate the maximum intensity (70 m/s). In summary, the Swin-Unet model tends to produce averaged estimates of TC intensity, suggesting a potential for further improvement, particularly in estimating the intensity of SSTY. Compared to the widely used DT, which exhibits systematic biases in estimating intensities of small/large TCs due to fixed cloud-ring criteria and subjective pattern recognition [50], our Swin-Unet model addresses these limitations through automated feature extraction based on self-attention mechanisms and dynamic storm morphology recognition (the rotation index and the dispersion index) that adapt to storm structural evolution. However, both methods struggle to estimate extreme intensities (> 60 m/s). Despite the observed tendency for the model to overestimate weak TCs and underestimate stronger TCs, this does not significantly compromise its overall estimation skill, as evidenced by an MAE of merely 3.71 m/s and a root mean square error (RMSE) of 5.05 m/s, which is considered acceptable within the context of disaster warning.

B. Monthly Distribution of Estimated Results

According to the statistics of the TC data in Section II-B, the TC season in the Western North Pacific is from June to November, especially from July to September, when TCs are most frequent. The TC records in these three months account for 65% of the annual total. Fig. 7 shows the estimated results from the Swin-Unet model during the period from June to November. It is obvious that except for June, the average values of the estimated TC intensity in other months (especially in July and August) are all less than that of $V_{\max}$. From the box plot, it is easy to find that the larger the estimated range of TC intensity, the greater the corresponding MAE. For example, the MAE exceeds 3 m/s from July to October, while, in June, the MAE is less than 3 m/s. From June to September, the MAE of the model gradually increases and then gradually decreases from September to November. This is understandable because STY or SSTY are more active during these months, leading to a certain degree of bias in the model's estimations. Therefore, the estimation performance of the Swin-Unet model is correlated with the active period of TCs.

C. Spatial Distribution of Estimated Results

Fig. 8 presents the spatial distributions of TC activity and the corresponding TC intensity estimation results. Fig. 8(a) shows that TC activity is concentrated over the subtropical ocean between 13°N and 30°N. This region exhibits a partial overlap with the areas of high MAE (> 2.4 m/s) in Fig. 8(b), suggesting that the Swin-Unet model also demonstrates considerable bias in estimating TC intensity within these active TC basins. This bias may stem from the complex air–sea interactions in this region, which make TC intensity inherently difficult to estimate. Fig. 8(c) reveals a systematic negative bias over the South China Sea and east of the Philippines (13°N – 25°N), indicating that the Swin-Unet model tends to underestimate the intensity of TCs at lower latitudes. As this region is also a primary TC genesis zone, this implies that the model underestimates the intensity of initial vortices. In addition, Fig. 8(c) shows sporadically distributed areas of high negative bias (< −9.6 m/s), which coincide with regions of low TC frequency. A comparison between 8(a) and (d) indicates that while TC activity is more prevalent west of 130°E compared to the east, the phenomenon of TC intensity overestimation occurs more frequently in the region east of 130°E. Overall, the model shows higher accuracy (lower MAE) in high-TC-frequency regions, e.g., the main development region of the Western North Pacific. Conversely, larger errors persist in regions with sparse TC records. While this pattern suggests that limited training samples in these low-activity regions might be a key driver of increased errors, current analysis cannot exclude the influences of unique regional meteorological conditions. For example, TCs here may be in different lifecycle stages (e.g., extratropical transition or land interaction) or subject to distinctive environmental conditions that the model has not adequately learned to capture due to limited meteorological data.

V. Discussion

A. Ablation Study

Since the Swin-Unet model incorporates storm morphology recognition (rotation index L_R and dispersion index L_D) as part of its loss function, it is essential to conduct an ablation study (component-impact analysis) to quantify the impact of both indices on the model's performance. Here, we configure four groups of control experiments and use the cross-validation method (similar to Section IV-A) to analyze the results. Note that despite the exclusion of the L_R or L_D from some experimental schemes, we still calculate their correlation coefficients with typhoon intensity from the pertinent feature maps. The results are listed in Table I. It can be found that when using the original loss function (L_{MAE}), the Swin-Unet model obtains an MAE of 3.92 m/s and an RMSE of 5.46 m/s. Although the

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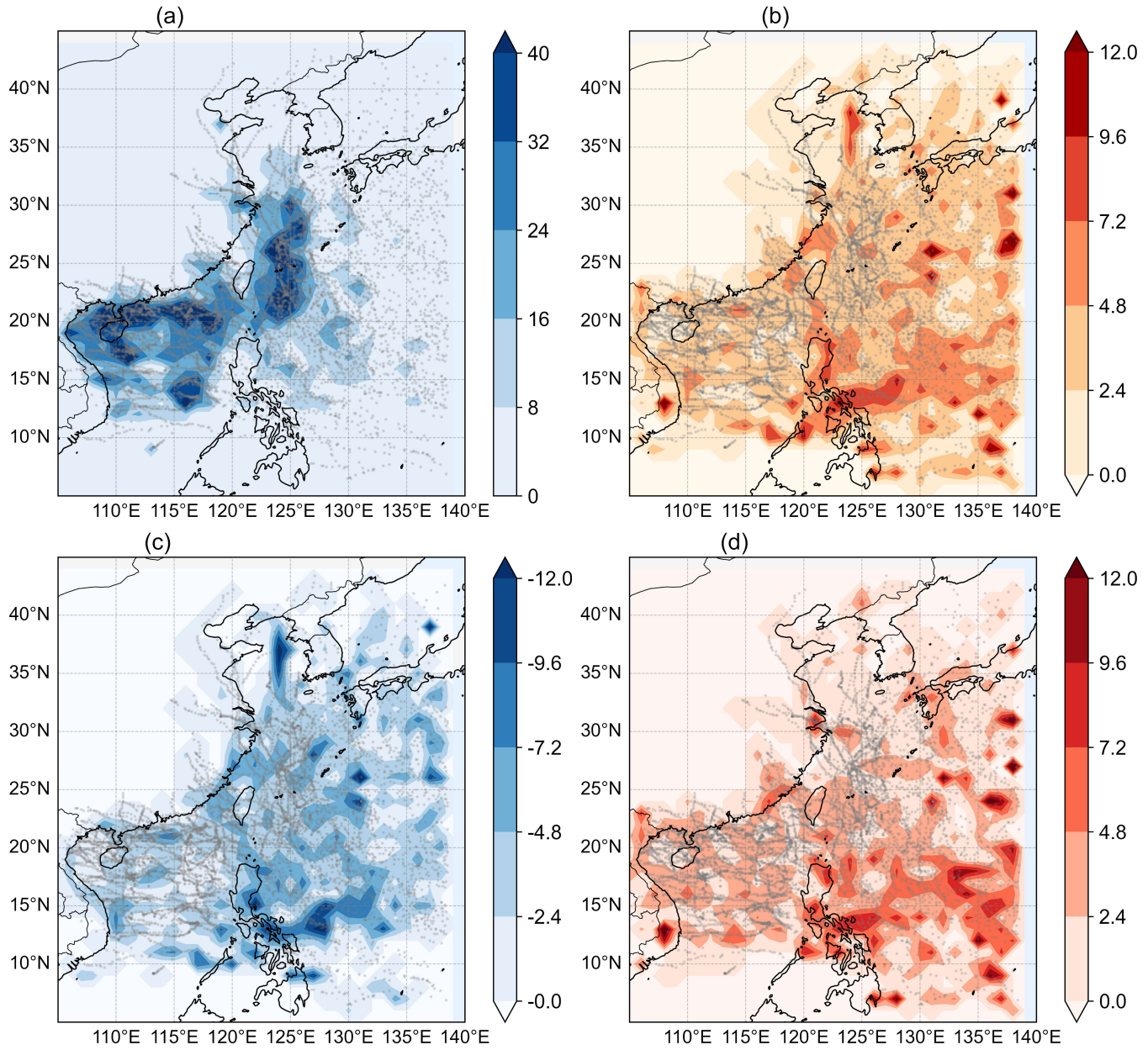


Fig. 8. Spatial distribution of TC intensity estimations over the region (105°E ~ 140°E, 5°N ~ 45°N). Results are visualized as 1° × 1° grid heatmaps. (a) Storm frequency (storm locations from best-track data, shown as gray dots). (b) MAE. (c) Negative bias. (d) Positive bias.

TABLE I

RESULTS OF ABLATION STUDIES THAT USE DIFFERENT LOSS FUNCTION SCHEMES; "CORR" MEANS THE CORRELATION COEFFICIENT BETWEEN THE ESTIMATED INTENSITY ($V_{\max}$) AND L_R OR L_D

Loss Function	MAE	RMSE	Corr (V_{max}, L_R)	Corr (V_{max}, L_D)
L_{MAE}	3.92	5.46	0.74	0.05
$L_{MAE} + L_R$	3.77	5.20	0.95	-0.51
$L_{MAE} + L_D$	4.04	5.49	0.95	0.10
$L_{MAE} + L_R + L_D$	3.71	5.05	0.88	-0.52

correlation coefficient between $V_{\max}$ and L_R reaches 0.74, the correlation coefficient between $V_{\max}$ and L_D is only 0.05, indicating that the model fails to capture the cyclone's structure. Subsequently, based on the original loss function, L_R and L_D are introduced, respectively. It is obvious that the former loss function ($L_{\text{MAE}} + L_R$) significantly improves the estimation accuracy of TC intensity, with the correlation coefficient between the estimated intensity ($V_{\max}$) and L_R being 0.95, while that between $V_{\max}$ and L_D is −0.51. This suggests that this scheme skillfully describes TC's structures. However, the latter loss function ($L_{\text{MAE}}+L_D$) differs, as its MAE and RMSE both increase compared to the original loss function. Despite

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TABLE II
COMPARISON OF THE SWIN-UNET MODEL WITH SEVERAL EXISTING STUDIES

Method/model	Data/Channel	MAE (m/s)	RMSE (m/s)	Literature
ADT	IR, visible/PMW imagery	4.46	5.77	[51]
DAV	GOES-12, IR (10.7 um)	/	6.68	[52]
DAV-T	MTSAT, IR (10.7 um)	/	6.55	[19]
pGCNN-TC	TCIR benchmark	/	5.46	[53]
MLP	SHIPS predictor	4.31	5.87	[54]
2D-CNN, 3D-CNN	COMS MI, IR1, IR2, WV, SWIR	3.13 $\sim$ 4.98	4.28 $\sim$ 6.67	[37]
TCICENet	GMS, GEO, MTSAT, H-8, etc.	3.83	4.93	[39]
D-MINT and D-PRINT	microwave/infrared imagery	/	4.19 $\sim$ 4.29	[41]
Swin-Unet	SEC, CTT, TBB, PRE, HPF	3.71	5.05	Ours

**Statement: Cross-method performance comparison exhibits inherent inconsistency since testing datasets differ across studies. For truly fair and accurate comparisons, all estimating methods need to be tested using the homogeneous storm data from the same time periods.*

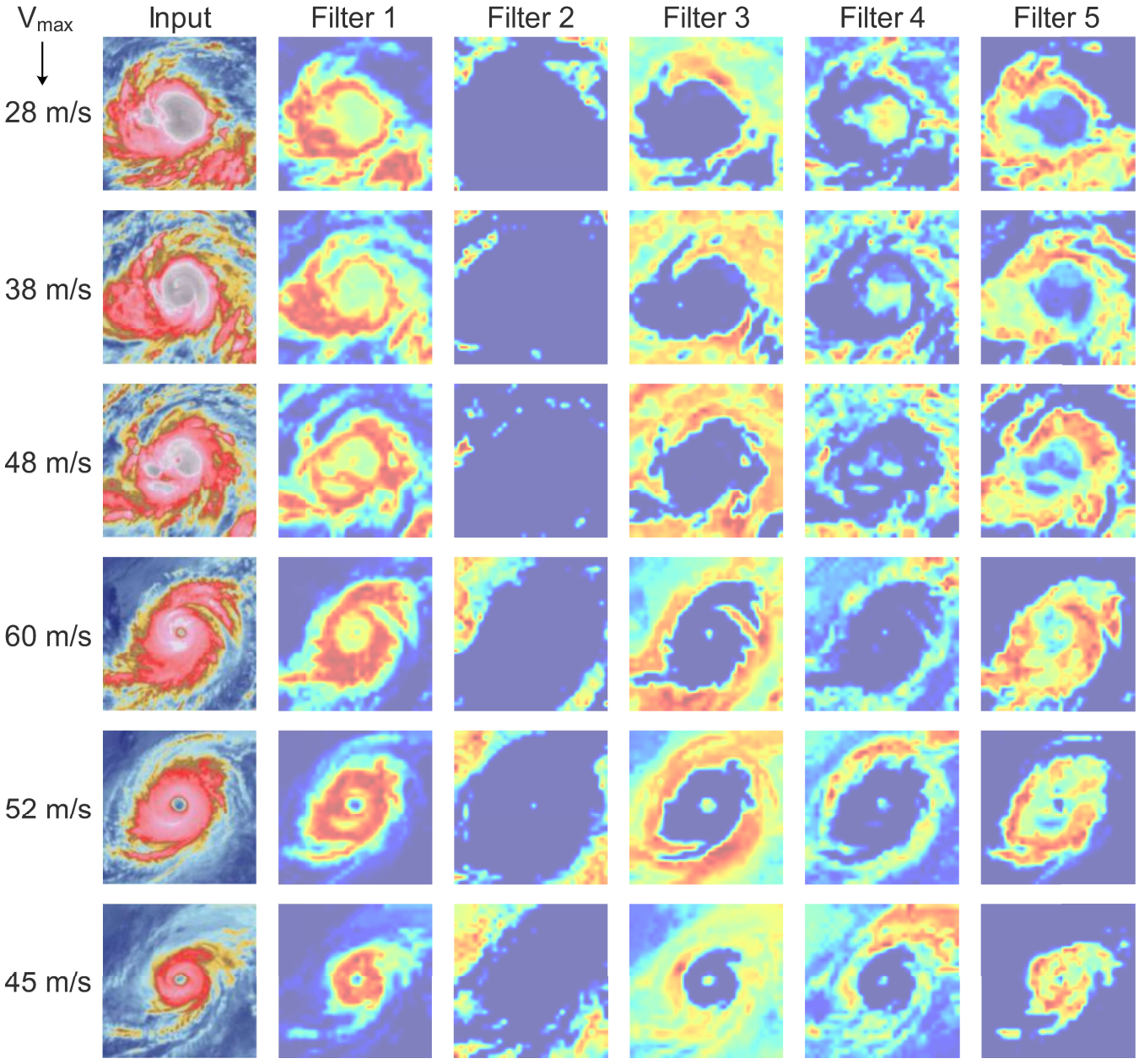


Fig. 9. CAM map of typhoon "Surigae" from one of the convolution layers in the Swin-Unet model.

the correlation coefficient between $V_{\max}$ and L_R attaining a high value of 0.95, the correlation coefficient between $V_{\max}$ and L_D fails to demonstrate a notable enhancement. Finally, upon incorporating L_R and L_D functions into the original loss function ($L_{\mathrm{MAE}} + L_R + L_D$), this scheme results in the minimal MAE of only 3.71 m/s and the minimal RMSE of only

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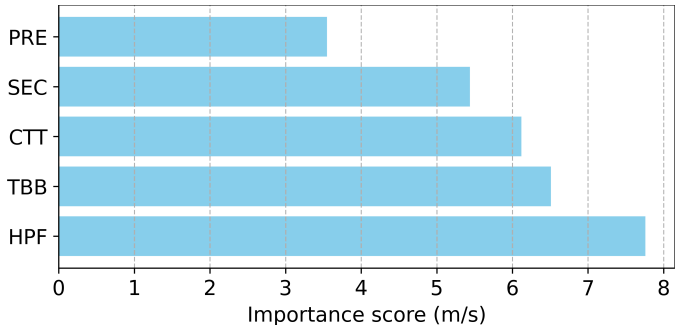


Fig. 10. Feature important scores of five satellite cloud products.

5.05 m/s. Moreover, the correlation coefficients between $V_{\max}$ and L_R, as well as L_D, are 0.89 and −0.52, respectively. The above results demonstrate that introducing either L_R or L_D improves the model's ability to characterize the rotational state of cyclones, but its ability to represent the dispersion state is quite limited. The model achieves a balance between estimating TC intensity and describing TC morphological characteristics only when both L_R and L_D are incorporated into the original loss function. Indeed, this phenomenon is understandable; even when most cloud systems congregate without developing a cyclone-like rotational pattern, they inherently fail to progress into TCs. Therefore, the Swin-Unet model in this study has a higher dependence on the rotation index L_R and a relatively lower dependence on the dispersion index L_D.

Here, we investigate the Swin-Unet model's advantages in estimating TC intensity from its architecture. First, the model uses the Swin transformer as its encoder to extract features from the input satellite imagery. The Swin transformer captures medium- or long-range dependencies within the satellite imagery through a self-attention mechanism, enabling a more comprehensive extraction of global features from the inputs. Second, in the decoding process, the model performs upsampling and feature fusion on the feature maps from the encoding process. This allows the model to identify features at multiple spatiotemporal scales within the satellite imagery while filtering out redundant ones. In addition, the model adopts skip connections between the encoder and decoder processes to combine features of different resolutions, enhancing the model's ability to represent detailed textures in satellite imagery. Third, the model incorporates both rotation index and dispersion index as part of its loss function, which not only optimizes the model's training process but also improves the model's ability to represent TC structures.

B. Comparison With Existing Studies

Here, we conduct a comprehensive comparison of the Swin-Unet model with existing TC intensity estimation models (see Table II). The latest version of the ADT boasts many advancements compared to its previous versions, and it is operationally utilized by most TC analysis and forecasting centers [51]. However, the ADT depends on the comprehensive application of multisource infrared imagery and the objective analysis of the storm's convective structures. In the western North Pacific, the MAE and RMSE of the ADT model are 4.46 and 5.77 m/s, respectively. The deviation angle variance (DAV) measures the level of organization of axisymmetry of a TC with the DAV metric as an indirect estimate of the TC intensity from infrared imagery [52]. Due to the inherent sensitivity of deviation angle signals to asymmetric storm structures [19], the DAV model yields a relatively high RMSE of 6.68 m/s, exceeding that of ADT. Later, in the basics of DAV, an improved technique (DAV-T) incorporates various satellite data and retrains the algorithm parameters specifically for TCs over different basins [19]. Nonetheless, the improvement remains insignificant. There are also some DL-based models for TC intensity estimation. For instance, the regression neural network pGCNN-TC model [53], which introduces convective core features and rotational symmetry to alleviate estimated biases, achieves a lower RMSE value of 5.46 m/s compared to the DAV-T model. The multilayer perceptron (MLP) model, which included a special testing scheme known as "leave-one-year-out" cross-validation, could overcome the overfitting issue commonly encountered to some extent and achieved low MAE and RMSE values [54]. Moreover, three models based on CNNs, namely, 2D-CNN, 3D-CNN, and TCICENet, are utilized to analyze the relationship between multispectral geostationary satellite images and TC intensity. These models exhibited excellent performance in estimating TC intensity, achieving MAE values lower than 5 m/s [37], [39]. The D-MINT and D-PRINT model achieved the lowest RMSE (4.19 m/s) through its multitemporal processing architecture, which integrates consecutive satellite imagery from preceding time steps. Although the above DL-based models outperform the DAV and ADT, they extract nonlinear features solely through the blind stacking of multiple convolutional layers, neglecting not only the interaction among various channels but also the diverse spatial characteristics of storm morphology. Importantly, our Swin-Unet model is comprised of transformer blocks and self-attention mechanisms, significantly enhancing cross-window interaction capabilities, along with facilitating robust feature extraction. Therefore, the model yields the lowest MAE of 3.71 m/s, indicating that its performance is comparable to the above existing models.

C. Interpretability of the Model

Essentially, the Swin-Unet model constitutes a black box system employing an "end-to-end" decision-making process, which utilizes five satellite cloud products as inputs to derive the regression values for TC intensity. To better understand this process, we employ the "class activation mapping" (CAM) technique to visualize the relevant features within the hierarchical structure of the model, thereby exploring the model's underlying decision mechanism. Here, we select one of the convolution layers from the "Patch Partition" module in the Swin-Unet (see Fig. 3) to obtain the CAM map. Fig. 9 shows the CAM map of typhoon "Surigae" (No. 202102) at different stages throughout its lifetime. Intuitively, the CAM map is generated based on the varying pixel values of the raw input, mirroring the diverse morphologies and structures of the TC. This is because different filters (convolutional kernels) in the convolution layer assign various weights to

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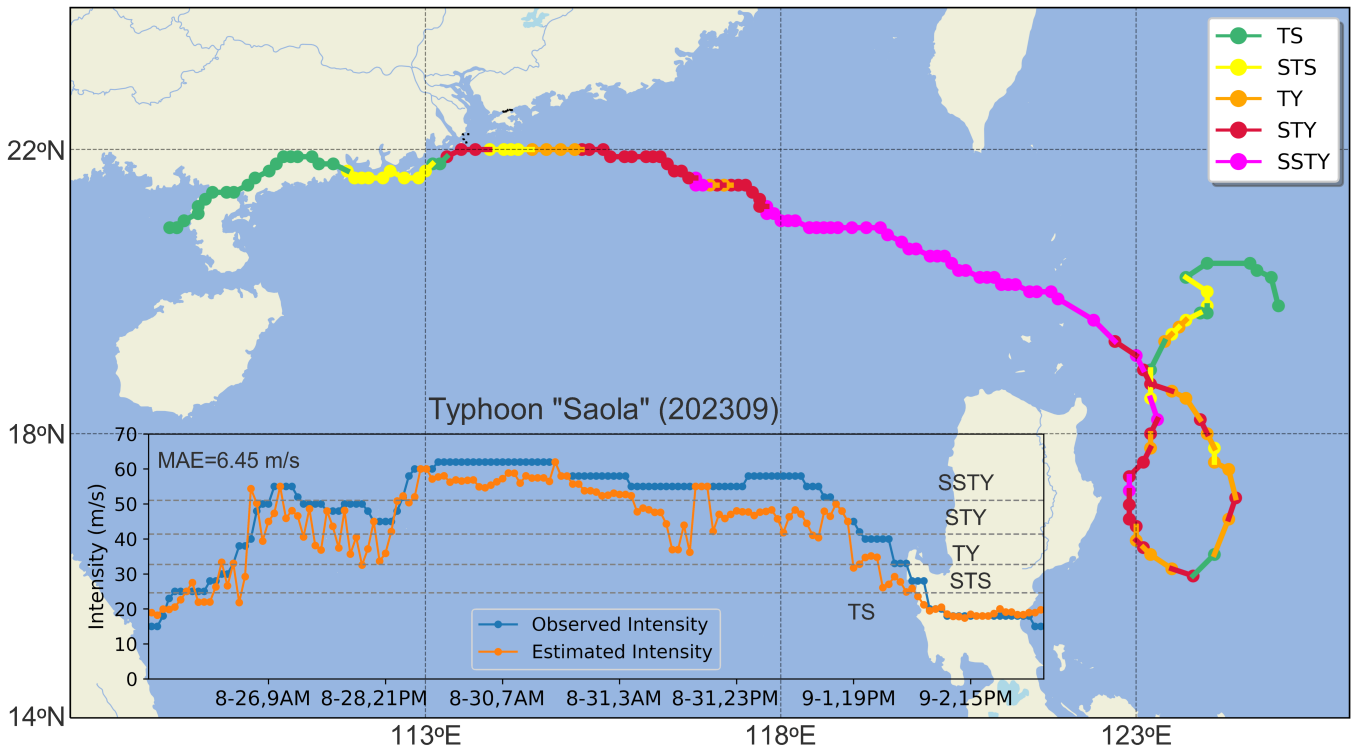


Fig. 11. Case study: the estimated results of the typhoon "Saola" (No. 202302). The storm track and colors are based on best track values.

these pixel values; in other words, they place varying emphases on describing the morphology and structure of TCs. According to the raw input (Fig. 9 only shows SEC), typhoon "Surigae" exhibits different spatial distributions in its cloud structure and rotational patterns at various intensities. For example, filter 1 effectively reveals the cloud system of the typhoon itself and the distribution of outer spiral rainbands. Furthermore, the density of these cloud systems can also reflect the intensity of convective activities within the typhoon, thereby enabling a reasonable assessment of the TC's intensity. In contrast, filter 2 focuses on describing the spatial distribution of outer cloud systems that exhibit a limited link with Surigae's convective activities. Nevertheless, these cloud systems still contribute a certain significance in estimating Surigae's intensity. Filter 3 pays more attention to regions with weaker convective activity of Surigae, including the outer spiral rainbands, middle spiral bands, and the typhoon's eyewall. Notably, as Surigae intensified from an initial wind speed of 28–60 m/s before weakening to 45 m/s, filter 3 accurately reproduced processes such as the formation and decay of Surigae's eye. Filters 4 and 5 seem to complement each other, as one focuses on the convective structure of Surigae and the inner spiral rainbands, while the other focuses on the outer spiral rainbands.

In addition, since the Swin-Unet model incorporates both transformer and Unet structures, the above filters not only focus on the cloud structure and rotational patterns of storms but also have different preferences for the five raw satellite cloud products. For example, the CAM maps in filter 1 are mainly activated by CTT and TBB, whereas those in filter 5 are mainly responsive to PRE and HPF (figures not shown). Consequently, while establishing a direct correlation between each CAM map and an individual cloud product remains challenging, the diversity of these CAM maps helps Swin-Unet capture complex aspects of TC intensity and further enhances the model's interpretability.

D. Feature Important Ranking

To assess the importance of each satellite cloud product in the Swin-Unet model, we set the corresponding imagery to zero and compared the model's output with that obtained using the raw imagery, thereby determining the importance of related satellite imagery. Fig. 10 shows the importance score of the five satellite cloud products. It is obvious that HPF has the highest importance score, reaching 7.76 m/s, suggesting that the Swin-Unet model in this study is most influenced by HPF in estimating a cyclone's intensity. Because HPF reflects the vertical distribution of water vapor in the atmosphere, it is conducive to revealing the characteristics and patterns of water vapor transport within cyclone areas. When water vapor transport is abundant, storms can gain more energy and moisture, which is favorable for maintaining or enhancing a cyclone's intensity. In addition, both CTT and TBB, which are characteristics reflecting "temperature," have importance scores greater than 6 m/s. In particular, CTT is closely related to upward airflow and convective activities within cyclone regions. A lower CTT usually indicates more intense energy release and dynamic processes within the cyclone, which contribute to the intensification of the cyclone. However, as shown by [55], while intense storms maintain substantial coverage of extremely cold cloud tops, spatial variability in CTT increases with eyewall maturation and inner-core convective organization. This reduces the direct correlation between mean CTT and storm intensity, a process

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linked to adjustments in inner-core dynamics [56]. Similarly, since the energy release and dynamic processes can lead to changes in thermal radiation intensity, the changes in TBB indirectly reflect cyclone intensity to a certain extent. In terms of SEC and PRE, their importance scores are both below 6 m/s, lower than those of HPF, CTT, and TBB. Apparently, the two features mainly indicate convective activity locations within cyclones, without addressing internal thermal and dynamic processes. To summarize, the Swin-Unet model mainly focuses on the thermodynamic and dynamic features intimately tied to cyclone convective activities, explaining the higher importance scores assigned to them.

E. Case Study

Here, we present a representative typhoon case to examine the performance of the Swin-Unet model (see Fig. 11). Typhoon "Saola" formed in the east of the Philippines on August 24, 2023. Its intensity rapidly increased over the following 24 h, from 30 m/s at 15:00 on August 25 to 55 m/s on August 26. The typhoon's track took a significant loop before continuing westward and finally making landfall in southern Guangdong Province, China, where its intensity quickly weakened. It can be found that the Swin-Unet model effectively reproduces the intensity variation process of Typhoon Saola. In particular, the model successfully captures the rapid intensification phase of Saola. However, the estimation bias is most evident during Saola's mature stage. Especially when Saola reaches the SSTY level (> 51 m/s), the model's estimated intensity tends to fluctuate and, in some cases, underestimates the $V_{\max}$. However, with an MAE of 6.45 m/s, the model already achieves the desired effect for early disaster warning.

VI. Conclusion

This study introduces a modified version of the "Swin-Unet" DL model, which integrates FY-series satellite cloud imagery for TC intensity estimation. Considering the intimate link between TC intensity and cyclone structure, we incorporate the rotation index and the divergence index into the model, utilizing them as parts of the loss function. This improvement allows the model to deliver more objective and rational estimates. The experimental outcomes further suggest that the incorporation of the two indices indeed elevates the model's estimation accuracy to a certain degree, achieving an MAE of 3.71 m/s. Notably, the rotation index demonstrates a particularly pronounced refinement in the model's estimation, potentially due to its strong correlation with TC convective activity. Meanwhile, the analysis of feature importance shows that HPF, CTT, and TBB have higher importance scores, suggesting that the model mainly focuses on the thermodynamic and dynamic features closely related to cyclone convective activities. In summary, the Swin-Unet model shows great potential in serving as a technical resource for real-time TC intensity estimation in coastal regions and offering decision support for disaster preparedness and warning systems.

It is well-known that in DL modeling, the algorithm often determines the lower bound of model performance, while the data determine the upper bound. The Swin-Unet model in this study incorporates the Unet architecture, self-attention mechanism, and Swin transformer block to maximize feature extraction from FY-series satellite cloud products. Experimental results demonstrate that the model exhibits improvements compared to other existing DL models or the Dvorak-based technique, and it also demonstrates greater robustness in estimating TC intensity. Of course, it is worthwhile exploring whether the incorporation of additional physical mechanism constraints into the Swin-Unet model can enhance the accuracy of TC intensity estimation by more precisely describing the thermodynamic and dynamic processes within TCs. In addition, it is important to tackle the problem that the Swin-Unet model often underestimates the high-level TC intensities, especially when dealing with extreme or severe cases. On one hand, we can mitigate the "data imbalance issue" among the raw TC samples through the data augmentation technique. On the other hand, incorporating data from multiple sources is also a strategy. For example, other geostationary satellite cloud products, numerical model outputs, real-time ocean observation data, ground-based remote sensing meteorological data, reanalysis data, and so on can all be considered for further validating and improving the model. Therefore, drawing on the above DL modeling approach and the experience acquired from leveraging existing FY-series satellite data, future works promise great contributions to predicting the rapid intensification of extreme TCs, TC tracks, and other TC metrics.

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